A Novel Convolutional Tsetlin Architecture for Classification of Satellite Images Under Limited Supervision

Priya Thangaraj
Shounak Chakraborty
priyat180156@gmail.com
shounak@iiitk.ac.in
Indian Institute of Information Technology Design and Manufacturing
Kurnool, Andhra Pradesh, India

Indrajit Kalita
indrajit@bu.edu
Faculty of Computing & Data Sciences, Boston University
Boston, MA, USA

Abstract

Tsetlin Machines (TMs) provide propositional rule-based learning that is natively interpretable, yet their application to high-dimensional remote sensing imagery under limited supervision remains largely unexplored. This paper introduces, to the best of our knowledge, the first unified CNN–Tsetlin Machine (CNN–TM) pipeline for multi-modal satellite image classification in low-data conditions with an inherent explainability for diverse RS images - multispectral, hyperspectral and synthetic aperature radars, multi-spectral and hyper-spectral images. This foundational work opens up research in investigating the extremely light-weight Tsetlin machines for processing decisions on-board a remote sensing platform. A multi-scale convolutional front-end extracts rich spatial features, which are converted into a compact Boolean representation through structured binary encoding that captures relative intensity, texture heterogeneity, and scatterer strength. The TM then learns propositional clauses over this representation, enabling both classification and attribution from a single mechanism without any post-hoc approximation. Evaluated on four publicly available benchmarks—MSTAR-10 (Synthetic Aperture Radar (SAR)), EuroSAT (multispectral), SAMPLE-10 (simulated SAR), and Indian Pines (hyperspectral)—under a strict budget of 121 labelled samples per class, the proposed framework achieves 98.39% accuracy on MSTAR-10, outperforming eleven deep learning baselines. Clause-driven attribution maps are structurally consistent with each prediction by construction, satisfying the interpretability demands of safety-critical deployment.

CCS Concepts

• **Computing methodologies → Neural networks**.

Keywords

automatic target recognition, binary feature encoding, interpretable machine learning, limited supervision, SAR image classification, Tsetlin Machine

ACM Reference Format:
Priya Thangaraj, Shounak Chakraborty, and Indrajit Kalita. 2025. A Novel Convolutional Tsetlin Architecture for Classification of Satellite Images Under Limited Supervision. In *Proceedings of Enter the correct conference title from your rights confirmation email (Conference acronym 'XX).* ACM, New York, NY, USA, 7 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Spaceborne sattelites image the Earth continuously across a wide range of modalities—from high-resolution optical and multispectral data to SAR and hyperspectral cubes [15] — enabling applications as varied as crop monitoring and soil-moisture estimation [12], urban land-use mapping [4], disaster damage assessment [2], and maritime surveillance [10]. Across all these domains, there is a shared and growing need for automated geospatial classifiers that are accurate, robust, and interpretable under realistic operating conditions.

Satisfying that need has proven difficult. Classical approaches such as support vector machines, spectral angle mappers, sparse dictionary methods, and random forests—required hand-engineered features cafelly curated out of a training dataset well-distributed across the defined classes [19]. Deep CNNs removed the feature engineering bottleneck and brought substantial accuracy gains across optical, multispectral, and hyperspectral imagery [18], with transformers, multimodal fusion, and few-shot learners pushing the frontier further [13]. Yet two structural problems persist regardless of architectural advances. First, deep representations are opaque: activations are distributed across millions of parameters, and post-hoc methods such as GradCAM and LIME reconstruct explanations through computations decoupled from the prediction itself, offering no causal guarantee [8, 9]. Second, deep networks overfit under limited supervision: trained on small annotated sets, they latch onto sample-specific patterns —speckle in SAR, calibration drift in multispectral data, band artifacts in hyperspectral imagery—and generalize poorly to shifted test conditions [11]. In operational remote sensing, where annotation budgets are chronically constrained and decisions carry real consequences, neither opacity nor fragility is acceptable.

Satellite imagery sharpens both problems simultaneously. Labels are costly to obtain, yet the data is inherently high-dimensional,

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
Conference acronym 'XX, City, Country
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-XXXX-X/2025/XX
https://doi.org/XXXXXXX.XXXXXXX

placing classifiers squarely in the regime where deep networks overfit and their opaque errors are hardest to diagnose [1]. A CNN front-end partly alleviates this: its hierarchical, localised receptive fields suppress acquisition-induced variation and extract structurally consistent features. But a downstream classifier is still needed that generalises from few labels and makes its reasoning transparent by design—not as an afterthought. This work proposes such a framework by pairing a multi-scale CNN with the Tsetlin Machine (TM) [6], a rule-learning classifier that operates over Boolean inputs, learns compact propositional clauses from limited data, and traces every prediction directly to specific input features without post-hoc approximation. An investigation using Tsetline machines is needed at this time since there is no single deep CNN-based architecture for classification across diverse RS domian like in SAR, multi-spectral and hyperspectral imagery. Also, this manuscript provides a foundation to Tsetlin architectures which are extremely light-weight and can be on-boarded to RS machinery like satellites and drones for a real-time analysis of the images. [16]

To the best of the our knowledge, a CNN–TM–based architecture has not yet been explored for the geospatial classification of remote sensing images under conditions of limited ground truth availability. Motivated by this, the present work investigates a convolutional TM framework for classifying satellite imagery across multiple domain, including SAR, multispectral, and hyperspectral data, while maintaining interpretability and robustness under limited supervision. The principal contributions are:

- **Multi-scale CNN feature extraction** with parallel branches at multiple kernel sizes captures complementary spatial structure across scales.
- **Sensor-invariant binary encoding** converts CNN activations into a compact Boolean vector via spatial partitioning and multi-threshold binarisation, requiring no modality-specific tuning.
- **Clause-driven interpretation with fidelity guarantees** derives attribution maps from the same clause set that governs each prediction, unlike decoupled post-hoc methods.
- **Cross-modal evaluation under limited supervision** that shows consistent gains over deep learning baselines implemented over SAR, multi-spectral and hyperspectral benchmarks.

2 Proposed Methodology

As shown in Figure 1, the proposed pipeline is structured as a sequence of three connected stages that link feature extraction with interpretable decision-making. The multi-scale CNN first processes the input image to produce a spatial feature representation $\mathbf{Z}$, capturing patterns at different levels of detail through parallel convolutional branches. This feature volume is then converted into a compact Boolean representation $\mathbf{b}$ by the structured binary encoder, which retains key spatial and relative characteristics while reducing sensitivity to variations in absolute values. The TM operates on this binary input to perform classification through rule-based learning, and at the same time generates a spatial attribution map $\mathbf{H}$ that reflects the basis of its decision. As illustrated in Figure 1, these components work in a tightly connected manner, allowing the model to move seamlessly from representation learning to transparent reasoning. Further details of each stage are provided in the following subsections.

Table 1: Structured Binary Feature Encoding Scheme

Bit	Condition	Type	SAR Interpretation
f_1	$\mu_{\text{rel}}>0.8$	Intensity	Mildly above channel mean
f_2	$\mu_{\text{rel}}>1.0$	Intensity	Clearly above channel mean
f_3	$\mu_{\text{rel}}>1.2$	Intensity	Dominant scattering centre
f_4	$\sigma_{\text{rel}}>0.8$	Texture	Mildly heterogeneous
f_5	$\sigma_{\text{rel}}>1.0$	Texture	Multi-bounce or clutter
f_6	cv>0.5	Contrast	Distributed scattering
f_7	cv>1.0	Contrast	Dihedral or shadow edge
f_8	$m_{ij}>\mu_c$	Peak	Notable point scatterer
f_9	$m_{ij}>\mu_c+\sigma_c$	Peak	Strong point scatterer

2.1 Input Preprocessing and Multi-Scale CNN

Each input is normalised to $[0, 1]$ and, if single-channel, replicated across three channels so that independent filter banks can be applied without introducing or discarding information. Three parallel convolutional branches with kernel sizes $k_1 < k_2 < k_3$—each zero-padded to preserve spatial dimensions—are applied simultaneously and their outputs concatenated channel-wise. Four sequential Conv→BN→ReLU blocks refine this representation: max-pooling after the first three blocks progressively halves spatial resolution while doubling channel depth, and the final block applies dropout at rate p to limit memorisation of sample-specific noise. The resulting feature volume is $\mathbf{Z} \in \mathbb{R}^{C\times H\times W}$, as shown in Figure 2. Kernel k_1 captures fine structural edges, k_2 mid-scale texture, and k_3 coarse intensity gradients.

2.2 Structured Binary Encoding

The encoder converts $\mathbf{Z}$ into $\mathbf{b}$ while retaining three physically meaningful image properties: relative scattering intensity, texture heterogeneity, and peak activation strength.

Each channel $\mathbf{Z}^c \in \mathbb{R}^{H\times W}$ is partitioned into a non-overlapping $G\times G$ grid of $s\times s$ patches $\mathbf{P}^c_{ij}$. For each channel c, a channel-wide mean μ_c and standard deviation σ_c are computed; per-patch statistics are then normalised relative to these channel references:

$$\mu_{\text{rel}} = \frac{\mu_{ij}}{\mu_c + \varepsilon}, \qquad \sigma_{\text{rel}} = \frac{\sigma_{ij}}{\sigma_c + \varepsilon}, \qquad \text{cv} = \frac{\sigma_{ij}}{\mu_{ij} + \varepsilon} \tag{1}$$

where μ_{ij}, σ_{ij}, m_{ij} are the patch mean, standard deviation, and maximum, and ε prevents division by zero. Relative normalisation renders all features invariant to inter-branch scaling differences. Each descriptor is then quantised by multi-threshold binarisation as defined in Table 1: three levels on μ_{rel} (bits f_1–f_3), two on σ_{rel} (f_4–f_5), two on cv (f_6–f_7), and two on patch maximum m_{ij} (f_8–f_9) with adaptive thresholds μ_c and $\mu_c+\sigma_c$. Each patch yields a B-bit vector; concatenating over all G^2 patches across all C feature map channels gives $\mathbf{b}$ of dimension $C \times G^2 \times B$, illustrated in Figure 3.

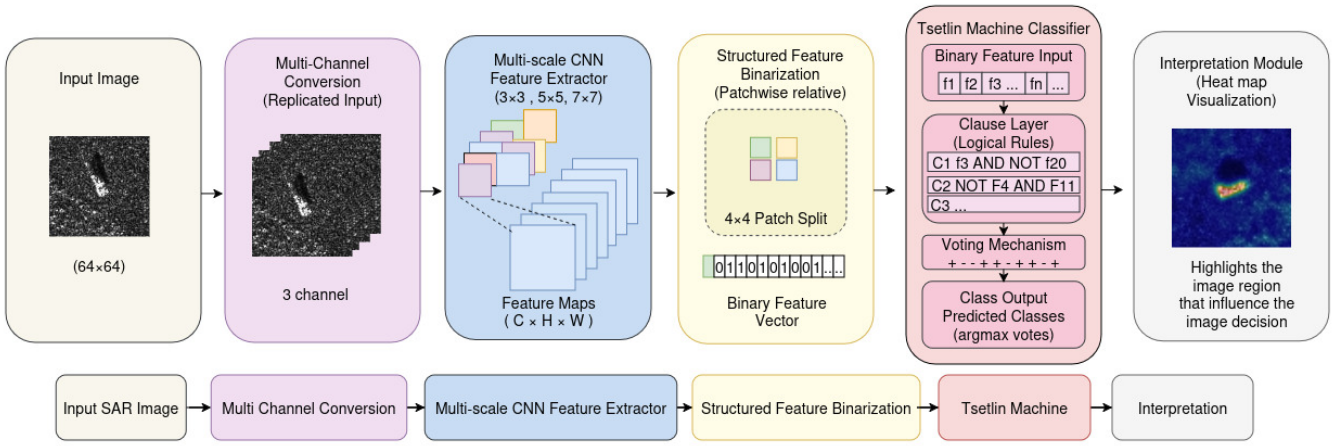


Figure 1: End-to-end CNN–TM pipeline. The input image is replicated to three channels, processed by the multi-scale CNN to produce Z, transformed into the Boolean vector b via structured binarisation, classified by the TM, and spatially attributed through clause-driven heatmap construction.

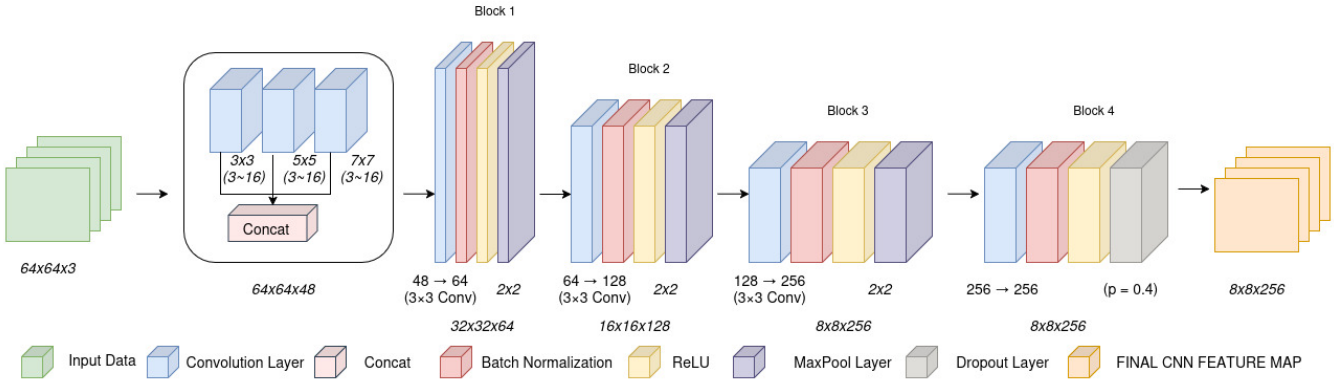


Figure 2: Multi-scale CNN architecture. Three parallel branches with kernels $k_1/k_2/k_3$ are concatenated and refined by four Conv-BN-ReLU blocks with progressive max-pooling, yielding $\mathbf{Z} \in \mathbb{R}^{C\times H\times W}$.

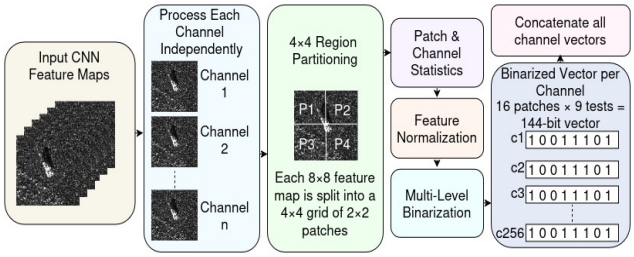


Figure 3: Structured binary encoding. Each channel of Z is divided into a $G\times G$ patch grid; relative descriptors $(\mu_{\text{rel}}, \sigma_{\text{rel}}, \mathbf{cv})$ are computed and multi-threshold binarised to produce a B-bit vector per patch, which are concatenated to form b.

2.3 Tsetlin Machine Classifier and Interpretation

The TM [6] operates over $\mathbf{b}$ by learning conjunctive clauses—Boolean combinations of features and their negations. Clauses vote for or against each class, and the class receiving the highest net vote is selected. Three hyperparameters govern this process: clause count C_{TM} controls capacity; voting threshold T clips aggregate votes to $[-T, +T]$, requiring broad clause consensus before peak confidence is assigned; and specificity s regulates clause granularity. Weighted clauses further amplify reliable rules and suppress noisy ones. Feature negation allows the TM to encode the *absence* of a response, which is especially informative in SAR where the missing dihedral return in a given spatial region can be as diagnostic as its presence.

For a test sample $\mathbf{x}_n$, the set of active binary features is $\mathcal{A}_n = \{k \mid \mathbf{x}_n[k] = 1\}$. Each $k \in \mathcal{A}_n$ indexes a channel c_k and patch (i_k, j_k) in $\mathbf{Z}$. The corresponding normalised patch response $\mathbf{R}_k$ is

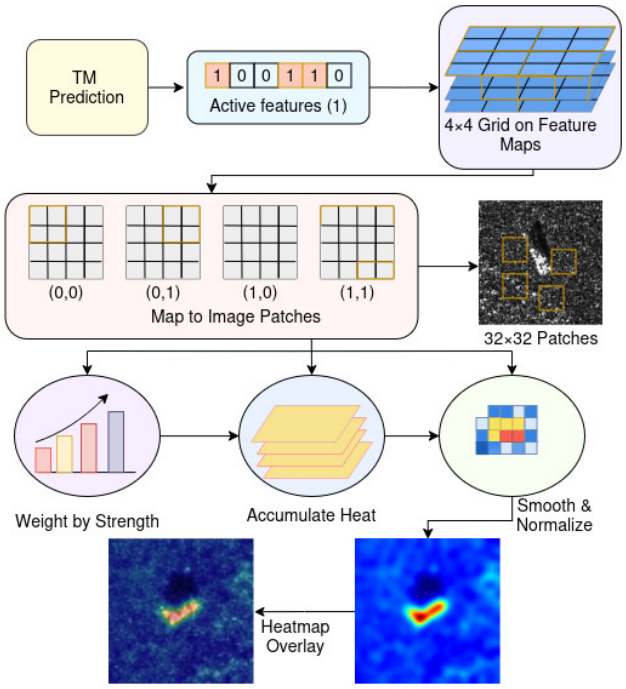


Figure 4: Clause-driven interpretation module. Active binary features are spatially localised in Z, accumulated with type-dependent weights (w_p for peak bits f_8/f_9, w_b otherwise), and Gaussian-smoothed to produce the attribution heatmap $\mathbf{H}_n$.

accumulated into a heatmap $\mathbf{H}_n$ (initialised to zero):

$$\mathbf{H}_n \mathrel{+}= w_k \cdot \frac{\mathbf{R}_k}{\max(\mathbf{R}_k) + \varepsilon}, \qquad w_k = \begin{cases} w_p & k \in \{f_8, f_9\} \\ w_b & \text{otherwise} \end{cases} \tag{2}$$

Peak-activation features f_8 and f_9 receive higher weight $w_p > w_b$ so that sparse but precise scatterer signatures are not overshadowed by the more numerous diffuse texture bits. $\mathbf{H}_n$ is then Gaussian-smoothed and bilinearly upsampled to the original image resolution, as illustrated in Figure 4. Crucially, $\mathbf{H}_n$ is derived from the same clause set that produced the prediction—no surrogate model or additional inference pass is required. This yields a structural fidelity guarantee that post-hoc methods such as GradCAM and LIME, whose explanations are formally decoupled from the classifier, cannot provide.

3 Experimental Setup

Four public benchmarks are evaluated at N_{train}=121 samples per class (Table 2). MSTAR-10 follows the standard cross-condition ATR protocol (train 17°, test 15°), posing the hardest generalisation challenge as acquisition conditions differ between splits. EuroSAT and SAMPLE-10 test transfer to optical reflectance and synthetic-to-real domains respectively, while Indian Pines probes whether Boolean CNN features retain discriminative value across ~200 spectral bands. No data augmentation or external pretraining is applied to any dataset, ensuring a clean measure of the pipeline's intrinsic capacity under limited supervision.

Table 2: Dataset Summary (N_{train}=121 per class; all images 64×64 pixels)

Dataset	Sensor	Cls.	Test	Task
MSTAR-10 [14]†	SAR (X-band)	10	2,426	ATR
EuroSAT [7]	Sentinel-2 RGB	10	Rem.	LULC
SAMPLE-10 [5]	SAR (simulated)	10	Rem.	ATR
Indian Pines [17]	AVIRIS (HSI)	10	Rem.	HSI

Table 3: Architecture and Hyperparameter Settings (MSTAR-10). Fixed parameters are identical across all datasets; ‡tuned independently per dataset via Optuna.

Parameter	Value	Parameter	Value
CNN		*Binary Encoder*	
Kernels $k_1/k_2/k_3$	3/5/7	Grid G / bits B	4 / 9
Output $C{\times}H{\times}W$†	256×8×8	Weights w_p / w_b	2.0 / 1.0
Dropout p	0.4	Stability ε	10^{-6}
LR η / decay λ	3e-4 / 1e-4	*Tsetlin Machine*	
Batch B_b / E_{cnn}	32 / 50	Clauses C_{TM}‡	4000
Smoothing ε_{ls}	0.1	Threshold T‡	2500
TM epochs E_{tm}	50	Specificity ρ‡	6.00

†Derived: 64×64 input, three 2×2 max-pools. ‡Optuna (100 trials, TPE): $C_{\text{TM}} \in [500, 5000]$, $T \in [10, 500]$, $\rho \in [2.0, 20.0]$.

The CNN is trained end-to-end with Adam using the settings in Table 3. All inputs are resized to 64×64 pixels; the output feature volume $\mathbf{Z} \in \mathbb{R}^{256\times8\times8}$ is derived from this fixed input size through three successive 2×2 max-pool operations and is therefore not a free parameter. An auxiliary linear head stabilises feature learning during CNN training and is discarded before TM training. The binary encoder partitions each channel of $\mathbf{Z}$ into a $G{\times}G$ grid, producing B=9 bits per patch from Table 1; peak bits f_8/f_9 receive elevated weight w_p to preserve sparse scatterer signatures. The three TM hyperparameters — clause count C_{TM}, voting threshold T, and specificity ρ (distinct from patch size s_p) — jointly control capacity, consensus requirement, and clause granularity; all three are tuned via Optuna (100 trials, TPE sampler) over the ranges in Table 3. Fixed parameters are identical across all four datasets; Optuna-tuned values are independently re-optimised per dataset to account for differences in modality, class separability, and feature distribution.

4 Results and Discussion

This section presents the performance of the proposed framework across multiple datasets, followed by a detailed analysis of its behaviour.

While cross-dataset performance is reported for all benchmarks, detailed analyses including confusion matrix evaluation, deep baseline comparison, and ablation studies are conducted on the MSTAR-10 dataset, as it represents the most challenging and widely used benchmark for SAR-based target recognition.

Table 4: Cross-Dataset Accuracy: CNN Baseline vs. CNN+TM

Dataset	CNN (%)	CNN+TM (%)
MSTAR-10	93.07	**98.39**
Indian Pines	32.05	**84.36**
SAMPLE-10	61.78	**100.00**
EuroSAT	80.74	**87.37**

4.1 Cross-Dataset Performance

Table 4 reports accuracy for the CNN baseline and the full CNN+TM pipeline across all four benchmarks. The proposed framework consistently surpasses the baseline on every modality. The most substantial gain occurs on Indian Pines (32.05% → 84.36%): structured binary encoding effectively captures relative spectral-spatial variation across ∼200 channels, and TM clauses generalise stably from the limited training pool. On MSTAR, 98.39% demonstrates that multi-scale features combined with rule-based learning reliably encode the SAR scattering characteristics that matter for ATR—structural edges, shadow boundaries, dominant reflectors—while the binary interface suppresses the speckle-driven overfitting that degrades purely gradient-trained models.The confusion matrix for MSTAR-10 (Figure 5) exhibits a predominantly diagonal structure, confirming that the proposed model achieves highly reliable class-wise discrimination. The limited misclassifications are concentrated among structurally similar vehicle classes, such as BMP2, BTR60, and BTR70, which share comparable scattering signatures and geometric layouts. This behaviour indicates that the model effectively captures dominant target-specific features while remaining robust to cross-condition variations, with residual errors arising from intrinsic inter-class similarity rather than instability in learning. Consistent gains on EuroSAT and SAMPLE-10 further confirm that the binary encoding generalises across coherent SAR, optical, and simulated imaging domains alike.

4.2 Comparison with Deep Learning Baselines

Table 5 benchmarks CNN+TM against eleven widely used architectures on MSTAR-10 dataset initialised with ImageNet pretrained weights and fine-tuned on MSTAR under the same N_{train} constraint. The proposed framework leads by at least 9% over the next best model. Two factors account for this margin. First, the CNN's hierarchical convolutions and dropout suppress acquisition-specific speckle, while multi-threshold binarisation provides a second regularisation stage by discarding the residual absolute activation magnitudes that survive feature extraction — preserving only ordinal structure and thereby preventing the TM from overfitting to sample-specific intensity variation. Second, the TM's propositional inductive bias—compact clauses over a discrete input space—is particularly well suited to limited-data regimes where models with millions of free parameters interpolate noise rather than structure.

4.3 Convergence and Ablation Studies

Figure 6 and 7 presents convergence trajectories and two ablation studies. The TM component converges monotonically within approximately 20 epochs, a consequence of its discrete automata-based update rule that eliminates gradient oscillation. Standard deep networks require up to 40 epochs before stabilising, exhibiting characteristic oscillatory behaviour under high dropout and small batch sizes. The multi-scale ablation confirms that the three-scale front-end consistently outperforms all single-scale and two-scale configurations. The patch grid ablation shows that the finest $G \times G$ configuration achieves the best accuracy, establishing that spatial

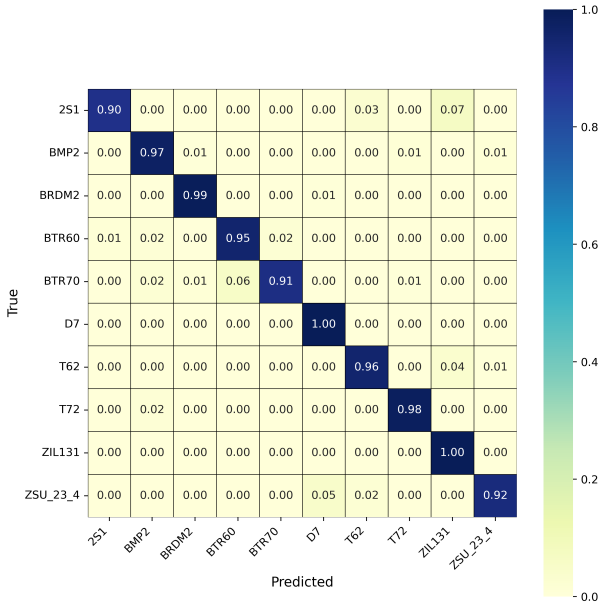


Figure 5: Confusion matrix for MSTAR-10. The strong diagonal dominance reflects robust class discrimination, while limited off-diagonal entries correspond to structurally similar targets with comparable scattering signatures.

Table 5: Deep Baseline Comparison on MSTAR-10 (N_{train}=121 per class)

Model	Accuracy (%)
Proposed (CNN+TM)	**98.39**
EfficientNetV2S	89.36
EfficientNetB0	87.92
ResNet50V2	85.48
EfficientNetB2	84.66
EfficientNetB1	83.87
ResNet101V2	83.09
ResNet152V2	82.27
MobileNetV2	57.57
NASNetMobile	15.26
MobileNetV3Large	11.42
MobileNetV3Small	11.29

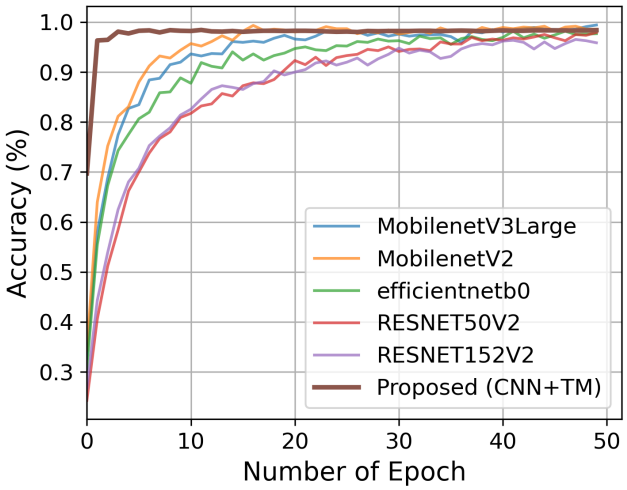
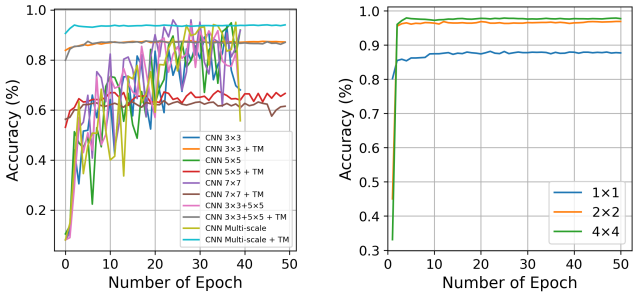
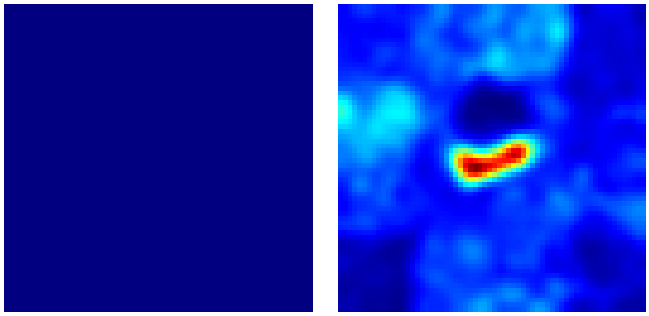


Figure 6: Training convergence of CNN+TM compared with deep baselines over E epochs on MSTAR-10.



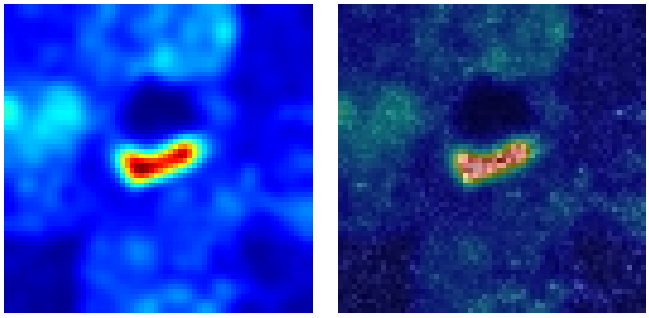
(a) Multi-scale **(b) Patch grid**

Figure 7: (a) Accuracy comparison of single-, two-, and three-scale front-ends with CNN and TM heads. (b) Effect of spatial patch grid resolution on MSTAR-10 accuracy.



(a) Standard binarisation **(b) Proposed encoding**

Figure 8: Attribution maps on MSTAR-10. Standard binarisation (left) yields a spatially diffuse response that fails to localise the vehicle. The proposed structured encoding (right) concentrates attribution on physically meaningful scattering regions. Both achieve equivalent classification accuracy; the difference lies entirely in attribution quality.



(a) Heatmap $\mathbf{H}_n$ **(b) Overlay on SAR image**

Figure 9: Spatial attribution for the 2S1 class (MSTAR-10). Hull corners and shadow boundaries—consistent with dominant X-band scattering mechanisms—are prominently highlighted, confirming that the heatmap is physically grounded.

locality within the feature map is essential for distinguishing vehicle classes that share similar global scattering statistics but differ in structural layout.

4.4 Interpretability and Spatial Attribution

To isolate the contribution of structured encoding to attribution quality, a reference system was built using standard 4-bit binarisation per feature channel [3], without the proposed patch-relative normalisation. This reference matches the classification accuracy of the full system at 98.39% on MSTAR-10, confirming that competitive recognition does not require structured encoding per se. However, as Figure 8 shows, its attribution map is spatially diffuse: because standard binarisation operates channel-wide without spatial partitioning, the resulting bits carry no reliable positional meaning and the interpretation module cannot recover coherent spatial attributions from them.

The proposed patch-relative normalisation preserves local spatial statistics within each channel, allowing active features to be traced back precisely to their originating patch in $\mathbf{Z}$. As Figure 9 confirms, the highlighted regions correspond to hull corners and shadow boundaries—the dominant X-band scattering mechanisms—and since $\mathbf{H}_n$ is assembled from exactly the features that drove the prediction, this agreement is structurally guaranteed rather than post-hoc approximated.

5 Conclusion

This paper presented a CNN–TM framework that addresses the three core challenges of operational remote sensing classification: limited labeled data, susceptibility to sensor noise, and decision opacity. By decoupling feature extraction from decision-making, the pipeline lets a multi-scale CNN capture rich spatial representations while the TM handles generalisation and interpretability through compact propositional rules. Structured binary encoding bridges these two components, converting continuous CNN activations into channel-normalised Boolean vectors that retain relative intensity, texture, and peak scatterer information while discarding

absolute magnitudes that drive overfitting. Attribution heatmaps are derived from the same clause set governing each prediction, providing a structural fidelity guarantee that post-hoc methods cannot match. Evaluated across four sensing modalities under a uniform 121-sample-per-class budget, the framework achieves 98.39% on MSTAR-10, outperforming all evaluated baselines including EfficientNetV2S (89.36%) and ResNet50V2 (85.48%). Future directions include polarimetric and complex-valued SAR extensions and online TM adaptation to non-stationary target distributions. This foundational work paves the way for developing extremely lightweight Tsetlin machines capable of performing on-board decision-making in remote sensing systems.

Acknowledgments

References

[1] Muhammad Ahmad, Shahzad Shabbir, Swalpa Kumar Roy, Danfeng Hong, Xin Wu, Antonio Plaza, and Jocelyn Chanussot. 2024. A Comprehensive Survey for Hyperspectral Image Classification: The Evolution from Conventional to Transformers and Mamba Models. *arXiv preprint arXiv:2404.14955* (2024). Under review at IEEE TGRS; pre-print available at https://arxiv.org/abs/2404.14955.

[2] Aaqib Ali, Mubashir Aziz, Ali Murtaza Rasool, Umair Ali, Muhammad Umar, and KMN Saquib Wani. 2025. Applications of remote sensing technologies to assess ground displacements and seismic hazards: Precision, scale, and insight. *Arabian Journal for Science and Engineering* 50, 20 (2025), 16357–16378.

[3] K Darshana Abeyrathna, Ole-Christoffer Granmo, Xuan Zhang, Lei Jiao, and Morten Goodwin. 2020. The regression Tsetlin machine: A novel approach to interpretable nonlinear regression. *Philosophical Transactions of the Royal Society A* 378, 2164 (2020), 20190165.

[4] Chengbin Deng and Zhe Zhu. 2020. Continuous subpixel monitoring of urban impervious surface using Landsat time series. *Remote Sensing of Environment* 238 (2020), 110929.

[5] Alexander W Denton. 2023. Data Augmentation for Synthetic Aperture Radar using Alpha Blending and Deep Layer Training. (2023).

[6] Ole-Christoffer Granmo. 2018. The Tsetlin Machine—A Game Theoretic Bandit Driven Approach to Optimal Pattern Recognition with Propositional Logic. *arXiv preprint arXiv:1804.01508* (2018).

[7] Patrick Helber, Benjamin Bischke, Andreas Dengel, and Damian Borth. 2019. EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 12, 7 (2019), 2217–2226.

[8] Adrian Höhl, Ivica Obadic, Miguel-Angel Fernandez-Torres, Haitham Najjar, Diego A. B. Oliveira, Zeynep Akata, and Xiao Xiang Zhu. 2024. Opening the Black Box: A Systematic Review on Explainable Artificial Intelligence in Remote Sensing. *IEEE Geoscience and Remote Sensing Magazine* (2024). doi:10.1109/MGRS.2024.3387709

[9] Ioannis Kakogeorgiou and Konstantinos Karantzalos. 2021. Evaluating Explainable Artificial Intelligence Methods for Multi-Label Deep Learning Classification Tasks in Remote Sensing. *ISPRS Journal of Photogrammetry and Remote Sensing* 173 (2021), 262–272. doi:10.1016/j.isprsjprs.2021.01.007 Cite as post-2020 ISPRS JPRS; verify author list against your reference manager.

[10] Jun Li, Yanqiu Pei, Shaohua Zhao, Rulin Xiao, Xiao Sang, and Chengye Zhang. 2020. A review of remote sensing for environmental monitoring in China. *Remote Sensing* 12, 7 (2020), 1130.

[11] Zhaokui Li, Hao Guo, Yushi Chen, Chenyang Liu, Qian Du, Zhiling Fang, and Yalin Wang. 2023. Few-Shot Hyperspectral Image Classification with Self-Supervised Learning. *IEEE Transactions on Geoscience and Remote Sensing* 61 (2023), 1–17. doi:10.1109/TGRS.2023.3234252

[12] Ranganath R Navalgund, Vivek Jayaraman, and Parth Sarathi Roy. 2007. Remote sensing applications: An overview. *current science* (2007), 1747–1766.

[13] Swalpa Kumar Roy, Ankur Deria, Danfeng Hong, Behnood Rasti, Antonio Plaza, and Jocelyn Chanussot. 2023. Multimodal Fusion Transformer for Remote Sensing Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* 61 (2023), 1–20. doi:10.1109/TGRS.2023.3286826 Art no. 5515620.

[14] Qian Song, Kefeng Ji, Yongping Zhang, Xiangwei Xing, and Huanxin Zou. 2016. Sparse representation-based SAR image target classification on the 10-class MSTAR data set. *Applied Sciences* 6, 1 (2016), 26.

[15] Devis Tuia, Konrad Schindler, Lorenzo Bruzzone, Utsav Bhatt, Gustau Camps-Valls, Xiao Xiang Zhu, Pedram Ghamisi, Bertrand Le Saux, Michele Volpi, and Gabriele Moser. 2021. Toward a Collective Agenda on AI for Earth Observation Data Analysis. *IEEE Geoscience and Remote Sensing Magazine* 9, 2 (2021), 88–104. doi:10.1109/MGRS.2020.3031080

[16] Svein Anders Tunheim, Lei Jiao, Rishad Shafik, Alex Yakovlev, and Ole-Christoffer Granmo. 2024. Tsetlin machine-based image classification FPGA accelerator with on-device training. *IEEE Transactions on Circuits and Systems I: Regular Papers* 72, 2 (2024), 830–843.

[17] Vani Vishwanath, K. Sreekanth, Jithu Prakash, Ananthi Rajendran, and G. Gopakumar. 2024. Hyperspectral patterns with deep learning for classification for Indian Pines. In *Proceedings of the 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*. IEEE, 1–7.

[18] Ke Xu, Ping Deng, and Hui Huang. 2022. Vision Transformer: An Excellent Teacher for Guiding Small Networks in Remote Sensing Image Scene Classification. *IEEE Transactions on Geoscience and Remote Sensing* 60 (2022), 1–15. doi:10.1109/TGRS.2022.3152566

[19] Xiao Xiang Zhu, Devis Tuia, Lichao Mou, Gui-Song Xia, Liangpei Zhang, Feng Xu, and Friedrich Fraundorfer. 2017. Deep learning in remote sensing: A comprehensive review and list of resources. *IEEE Geoscience and Remote Sensing Magazine* 5, 4 (2017), 8–36.